Paralyzed Veterans of America (PVA)

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1. Introduction

The dataset used was provided by the Paralyzed Veterans of America (PVA). PVA is a non-profit organization that provides programs and services for US veterans with spinal cord injuries or disease.

Our task was to analyze a sample of the results of one of PVA's recent fundraising appeals, containing 95 412 “lapsed” donors. This group is of particular interest to PVA as these individuals, who made their last donation to PVA between 13 to 24 months ago, represent an important group to PVA, since the longer someone goes without donating, the less likely they will be to give again.

With this project, we pretend to segment the lapsed PVA donors and create targeted marketing strategies for each group to recapture these former donors as this is a critical aspect of PVA's fundraising efforts.

2. Data Understanding and Exploration

Our initial dataset had 95 412 donors with 476 features for each donor and resulted from a combination of several sources. With such a big volume of observations and features, we will go through an overview of the whole dataset and select the features which we believe to be both relevant and workable (low number of missing values, coherent data, etc.).

When doing our initial exploration and manipulation of the data, we noticed that some “empty cells” were filled with a blank space which, depending on the data type of the feature, would not be counted as missing values as they should be. Moreover, we changed all the dates to “datetime” and all the possible features that could be transformed into binary (RECINHSE or CRAFTS, for instance) in order to increase their interpretability.

After this first overall look at our features and changes in datatypes, we decided to remove what we considered to be non-essential features: features that were obviously redundant with another feature; attributes that were not meaningful to our project objectives; variables that were not clearly defined or when the definition in the metadata mismatched the data and also ambiguous features, such as HOMEOWNR, where “unknown” status makes it impossible to understand if that donor does in fact owns a house, or not. We also noticed that in most of the features, the cell representing 0 is left empty (e.g., if the number of children = 0, the cell is empty, and the value is NaN). Hence, it is impossible to tell whether the data is missing or the cell represents 0. We made an assumption here and converted all the NaN values in these specific features to 0.

Following this feature removal, we proceeded to study which features were highly correlated: INCOME, WEALTH1 and WEALTH2, for example, add correlation values of 98% thus we decided to drop both latter ones.

Besides data correction and removal, feature engineering was also performed in order to further reduce data cardinality and improve the performance of our model: for example, since the metadata doesn’t define what “A” and “C” refers to in the column GENDER [(1)], we encoded those and “U” (Unknown) as NaN values. Then, we estimated their gender using TCODE. First, clear gender titles were reflected (e.g., Mr. versus Mrs.). Then, for military titles, we assumed them to be male, considering more than 80% of the US military comprises of males[1]. For Rabbi, M was used as female rabbis are extremely rare. Total of only 8 individuals required this assumption of male titles. Other job titles were left as NaN, as there

is no obvious gender ratio in the field or gender distinction in the title. As a result, we obtained the gender information of 97% of the observations. Lastly, females and males were encoded as 0 and 1, respectively. and we dropped the remaining NaN GENDER individuals.

However, even after this data cleaning, we still had almost 240 features in our dataset, which, with our computing power, would be highly difficult to evaluate. With that in mind, further feature reduction was needed and so we decided to group the remaining variables based on their "category" to ease the analysis and do further removal based on a correlation matrix and our focus on interest. These 4 categories were: donors' information; neighborhood (who lives there, where do they work, education features, wealth, what's the percentage of veterans); marketing and gifts. After some more feature engineering, data dropping, and using our common sense we finally arrived at our final 55 features which we think are the most relevant ones to design and solve our donor segmentation problem.

3. In-depth Analysis on Selected Features

From our selected features, 22 are non-marketing related and will be used to group our donors based on their characteristics. Below, we divided our non-marketing features in two tables: the first one with "original features" that were already given on the original dataset and the second table with features engineered by us. In each table, we are explaining what each feature is and, if any, what pre-processing was done to this point.

FEATURE (original)	DEFINITION	PREPROCESS
IC1	Median Household Income in hundreds	None
NUMCHLD	Number of children	NA filled with 0
INCOME	Household income	None
PEPSTRFL	Indicates PEP Star RFA Status	None
GOV_LOC	Percent Employed by Local Government	In the original dataset the feature it's called OEDC1
GOV_STA	Percent Employed by State Government	In the original dataset the feature it's called OEDC2
GOV_FED	Percent Employed by Federal Government	In the original dataset the feature it's called OEDC3
VET	Percent Adult Veterans Age 16+ on the neighbourhood	In the original dataset the feature it's called AFC4
MIL_ACT	Percent Adults in Active Military Service	In the original dataset the feature it's called AFC1

RAMNTALL	Dollar amount of lifetime gifts to date	None
NGIFTALL	Number of lifetime gifts to date	None
CARDGIFT	Number of lifetime gifts to card promotions to date	None
AVGGIFT	Average dollar amount of gifts to date	None
TIMELAG	Number of months between first and second gift (if it's blank, the donor only did 1 donation)	None
MAXRAMNT	Dollar amount of largest gift to date	None

FEATURE (engineered)	**DEFINITION**	**PREPROCESS**
First_month	How recent was the first gift (in months)	First month is a feature that results from subtracting the current date to the ODATEDW (origin Date; date of donor's first gift to PVA). This difference is then transformed in months.
Donation_semester	How many times did the donor donated per semester, on average, since the first donation	Donation Semester results from the division of NGIFTALL (number of lifetime gifts to date) by the first month variables, multiplied by 6 (so it's a semester value).
MoneyPerYear	How much money has been donated on average per year by the donor	Product of AVGGIFT and Donation_semester multiplied by 2 as in donation we are talking about semester and not year
IncomevsGift	How much Income the donor uses on average to donate to PVA every year.	Division between MoneyPerYear and Income
GENDER	Gender of donor (M= Male; F= Female; J = Joint Account, unknown gender)	see note (1) in chapter 2 for the preprocessing explanation
Age of the donor	Age of the donor	Built from feature "DOB" (date of birth)
Education	Education level of donor	Education is: • 1 if the donor studied less than 11 years • 2 if the door studied between 11 and 13 years

		• 3 if the donor studied between 13 and 15 years • 4 if the donor studied more than 15 years This discretization took into consideration of mandatory education years in the US (10 years) and the length of secondary and tertiary education.

3.1 Data Import and Assessment

As stated above, after our initial data cleaning, we have 55 features of which 22 are descriptive (describe the donor, the neighborhood where he/she lives, the donations made) and the remaining are marketing-related (features that directly show the donor's interest/taste and will be important when preparing a marketing strategy). One of our main concerns when choosing which features would better segment PVA's donors, was to pick features that had minimal numbers of missing data. DOMAIN and GENDER, for instance, don't have more than 3% of the observations with missing values which we considered to be insignificant given the size of the dataset. On the other hand, features like INCOME or Age have more than 20% of their bservations with missing values, which was an issue that we had to solve in order to draw meaningful conclusions.

The DOMAIN feature consists of two elements, urbanicity and socioeconomic status (SES). SES is measured differently for DOMAIN codes (1 is always the highest SES independently of the Urbanicity level, being 3 the lowest one for C, S, T and R; while when Urbanicity is U, the lowest SES is 4). Since we do not know which measure was used for the SES measurement and what is the difference between each level, we have decided to only keep the Urbanicity definition of this feature (the 1st byte of the DOMAIN) and drop the SES information. This new feature is called URBANICITY and it will have the same code as DOMAIN. For the neighborhood income level, we will solely use IC1.

3.2 Data Cleaning

The main objective of data cleaning is to identify and remove errors and duplicates in the data, in order to create a reliable dataset. This will improve the quality of the dataset for analytics and enables accurate decision making. With this in mind, we searched the data for outliers, missing values, did a coherence check and scaled the data.

3.2.1 Outliers

At first, to look for outliers, we used a local outlier factor (LOF, see reference [2]). According to its documentation on sklearn, LOF considers as outliers the samples that have a substantially lower density than their neighbors.

In order to perform this method, we first scale the GIFT variables (NGIFTALL, RAMNTALL, AVGGIFT, Donation_Semester) using a standard scaler (standardizes features by removing the mean and scaling to unit variance, see reference [3]). Then, using these scaled values, we performed PCA with 2 principal components.

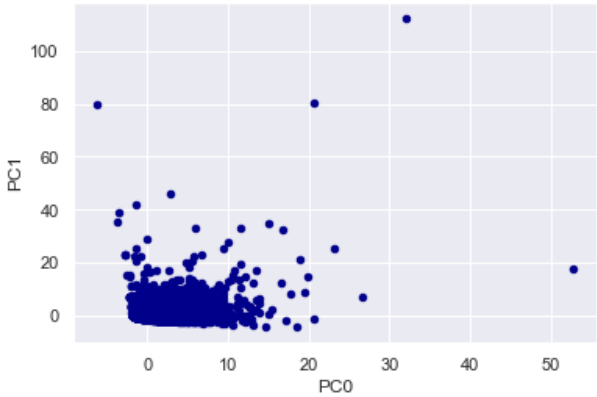

Fig. 1 GIFT variables PCA

Then, using LOF with 20 nearest neighbors, we decided to drop 1400 observations (1.5% of the total).

Also, we noticed that the variable AVGGIFT, average amount of dollars donate per gift, presents outliers, that we don't want to drop but we want to keep aside as high spending customers. Arbitrarily it has been decided to classify 150 dollars and above as a high amount of money per gift. The boxplots below show the dataset before and after cutting off those observations:

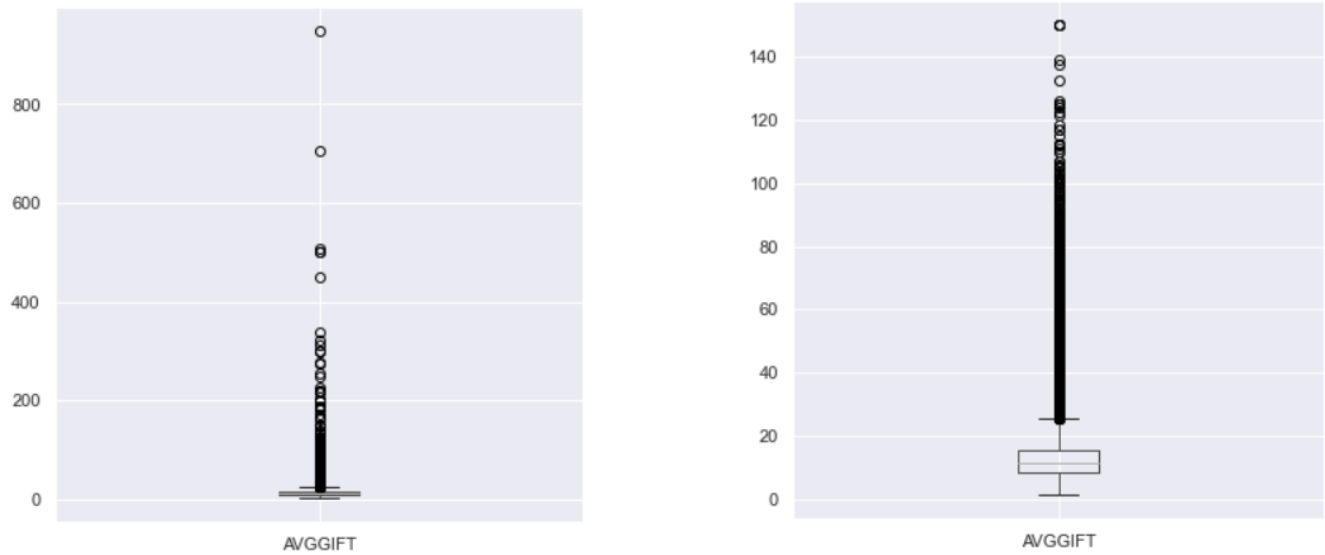

Fig. 2 AVGGIFT with and without outliers, respectively

3.2.2 Missing Values

Regarding missing values, we imputed the values to the variables which had NANs. In order to do so, we need firstly to scale them as we will use a distance algorithm, KNN. After scaling the data with missing values using a standard scaler, we imputed them using a KNN imputer with 300 nearest neighbors, using the common rule-of-thumb value of K=sqrt(n).

The variable Wealth was also imputed using KNN, with the same method as above.We utilized IC1, INCOME, Education, and EducationLevel based on Phi_K correlation. Despite some degree of correlation between Education and EducationLevel (=0.76), we used both variables, as they represent different characteristics: EducationLevel indicating the most common level of education achieved in the neighborhoor and Education estimating the level of education of the neighborhood as a whole. From a histogram, we noticed there is a linear relationship between the donor behavior and Wealth, i.e., there were more observations for increasing Wealth level. However, as the dataset consists only of lapsed donors, we could not definitively conclude whether wealthier individuals were more involved with charity overall.

3.2.3 Coherence Check

Coherence checks are used to verify that the data is coherent with reality. We will first drop all the observations which had people that had donated money before being born (277 observations) and we will also drop all observations which correspond to underage people (414 observations). We consider as underage a person with less than 16 years and we decided to remove them since we assumed that an underage person is not legally responsible for their finances and so cannot make any donations without adult supervision/consent.

3.2.4 Scaling

All continuous variables were scaled using a standard scaler. This is will be very important when using k-means, which will be discussed in the modeling chapter.

3.3 Data reduction

Data reduction is a process to reduce the volume of the data while maintining its integrity in order to make the data mining process more efficient. For that purpose, we used Principal Component Analysis (PCA).

3.3.1 Feature Engineering (PCA)

PCA stands for Principal Component Analysis (see reference [4] and [5]). PCA is “a dimensionality-reduction method that is often used to reduce the dimensionality of large data sets, by transforming a large set of variables into a smaller one that still contains most of the information in the large set” (see Zakaria Jaadi in References).

PCA comes at the expense of accuracy but makes the dataset smaller (therefore faster to run our models), easier to analyze and visualize. In summary, our goal when using PCA in this project was to reduce the number of variables of the dataset, while preserving as much information as possible.

In our project, we decided to run PCA on the GIFT variables so we can use its principal components on k-means (which we are going to discuss in the next chapter). The reason for choosing GIFT variables for PCA is that these variables "go together", meaning, they are strongly correlated: CARDGIFT, which we decided to drop for PCA, has a 0.9 correlation coefficient with NGIFTALL; MAXRAMNT is also highly correlated with AVGGIFT and since we are checking for the costumers' general behavior we decided to also drop it; finally, as RAMNTALL should be equal to AVGGIFT * NGIFTALL, we won't use it either for PCA. To come to the point, the variables we used for PCA were: First_month, AVGGIFT, NGIFTALL, Donation_semester, TIMELAG and MoneyPerYear. Below the correlation matrix of the gift variables:

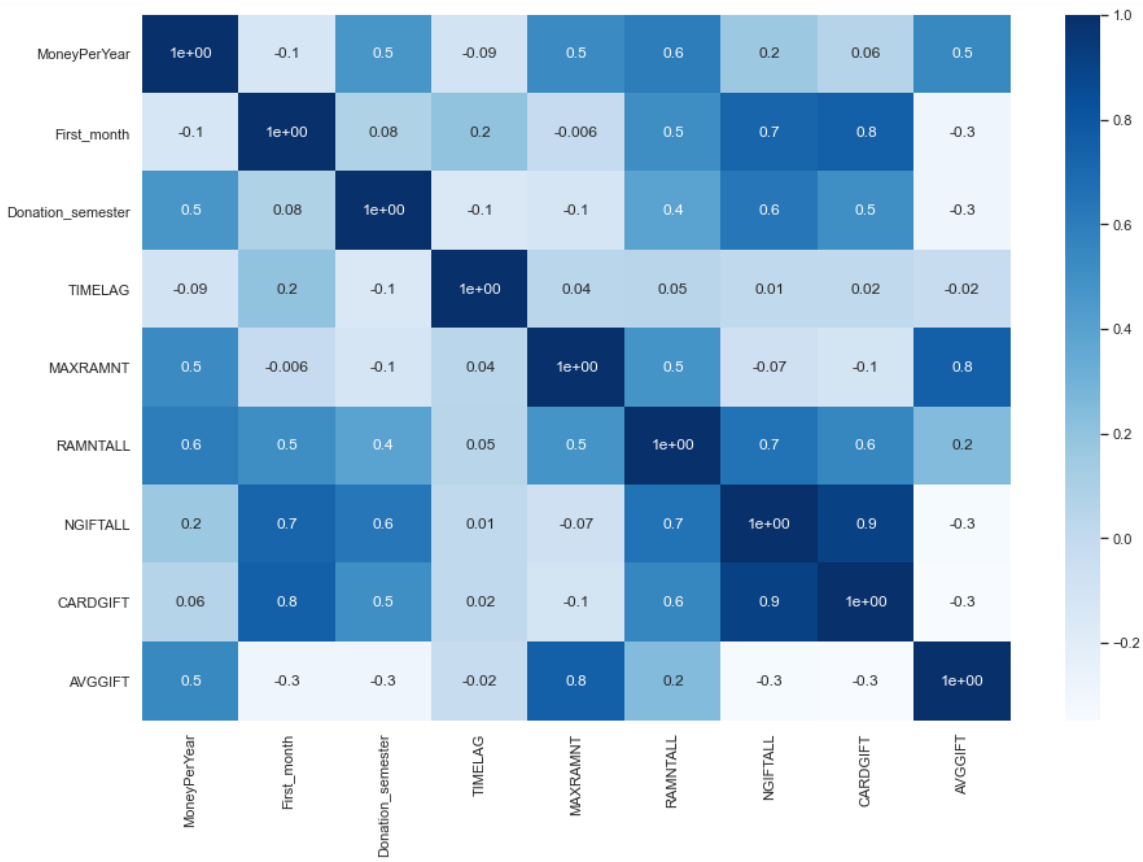

Fig. 3 Correlation matrix of the gift variables

From our PCA analysis we decide to choose the first 3 principal components (PCs), as they cumulative explain 84% of the variance in the data.

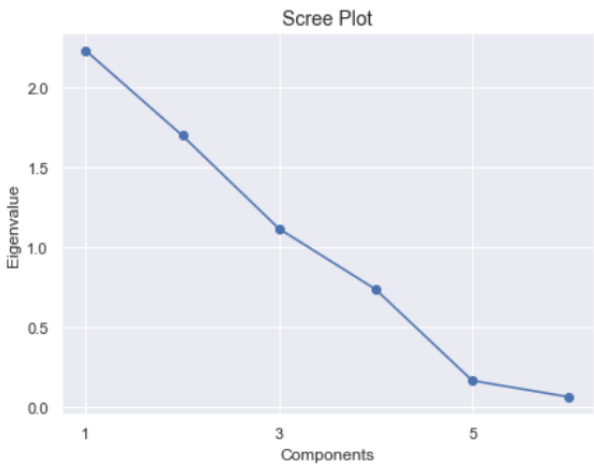

Fig. 4 GIFT variables PCA scree plot

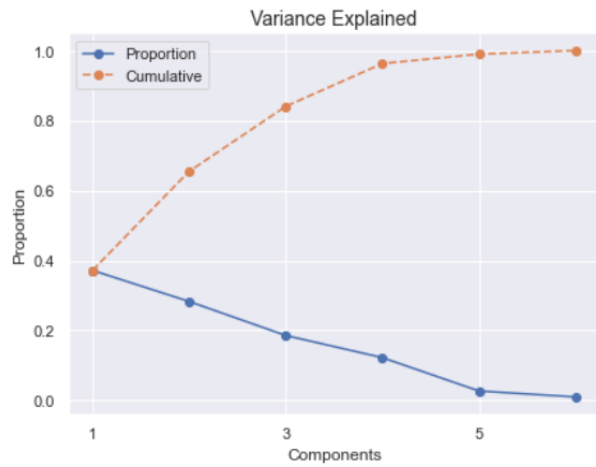

Fig. 5 GIFT variables PCA variance explained

The first linear combination is positively correlated with First_month, NGITALL, Donation_semester and negatively correlated to AVGGIFT. This means that the result of PC0 will be positively affected by the first variables and negatively affected by AVGGIFT. PC0 explains by itself 37% of the total variance of the 6 variables. PC1 is instead correlated with MoneyPerYear (which wasn't in PC0), AVGGIFT, Donation_semester and negatively correlated with First_month, meaning that the higher the first three variables the higher the PC1 value, on the contrary, the longer time ago the donator started helping PVA, the lower the PC1 will get.

	PC0	PC1	PC2
First_month	0.733934	-0.300007	0.393544
AVGGIFT	-0.561420	0.562039	0.474853
NGIFTALL	0.947503	0.126222	0.105936
Donation_semester	0.686884	0.544074	-0.263087
TIMELAG	0.053593	-0.342085	0.765094
MoneyPerYear	0.065406	0.928776	0.261285

Fig.6 Correlation matrix of PCs in terms of original variables

4. Modeling

We decided to use two main models for our donor segmentation: k-means for its simplicity and reliability, Gaussian Mixture Modelling (GMM), in opposition to k-means, for being a soft clustering method and, to visualize the results, t-distributed stochastic neighbor embedding (t-SNE). In addition, we experimented with k-prototype, which became popular due to its ability to cluster mixed data types.

4.1 k-means

K-means (see reference [6], [7] and [8]) is a partitional clustering algorithm. K-means algorithm classifies the data into K groups, where each group contains at least one point and each point belongs to exactly one cluster. Generically, the

criterion for a good partitioning is that objects belonging to the same cluster should be similar to one another but dissimilar to objects in other clusters.

In k-means, the object function used to assess the partitioning quality is the within-cluster variation, which is the sum of squared error between all objects in the cluster and its centroid. The centroid of a cluster is its center point. The centroid can be defined in various ways such as by the mean (like in k-means) or medoid (like in k-medoids) of the objects (or points) assigned to the cluster.

K-means is the most popular clustering algorithm as it is easy to understand and implement and also very efficient in terms of its time complexity. As a result, we decided to implement k-means as our first model.

For our k-means approach, we decided to start the model with 100 clusters and then agglomerate them via hierarchical clustering.

For the hierarchical clustering, besides the first 3 principal components discussed in the previous chapter, we used the variables age and Income. As you can see from the dendrogram below (Figure 6) 5 clusters seem to be a good fit for our problem. Using 3 clusters would also be possible, but with more than 90 000 observations, using 3 clusters would be too approximative.

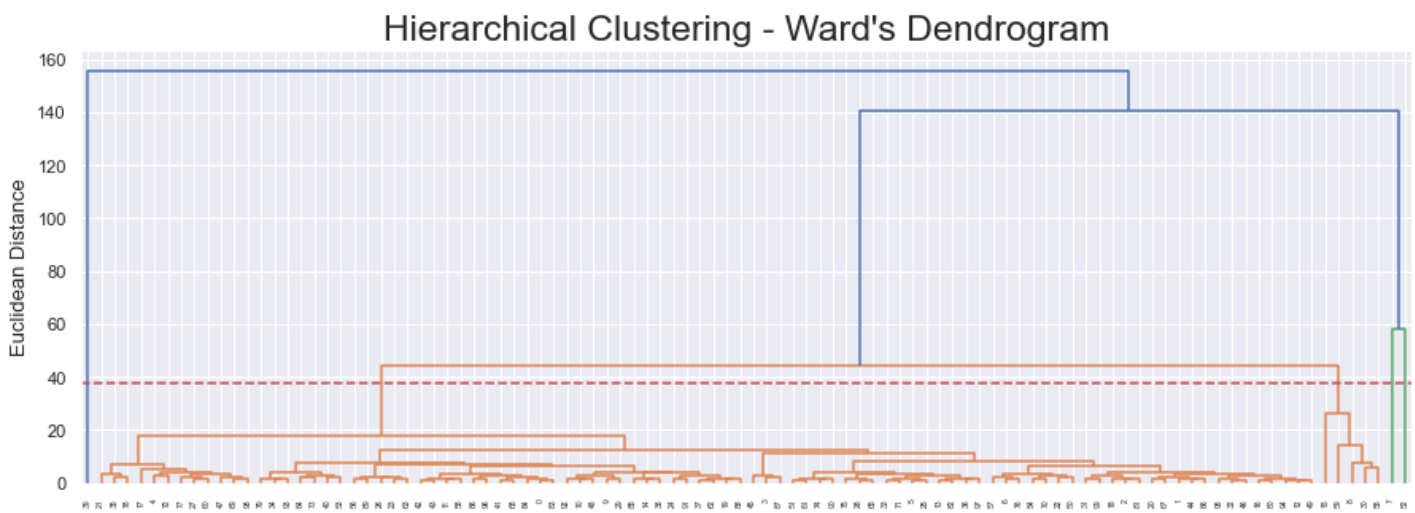

Fig.7 Hierarchical clustering – ward's dendrogram

And so, after analyzing the graph, we decided to proceed with 5 clusters when using the k-means algorithm.

```
lista_pca = df_pca[['PC0','PC1','PC2','Age', 'INCOME']]
```

```
number_clusters = 5
kmclust = KMeans(n_clusters=number_clusters, init='k-means++', n_init=15, random_state=1)
km_labels = kmclust.fit_predict(lista_pca)
```

Fig.8 K-means python code, using 5 clusters

We used “init = k-means++ “as this initializes the centroids to be (generally) distant from each other, leading to probably better results than random initialization. Also, “n_init = 15” allows to initialize KMeans 15 times and pick the best clustering in terms of Inertia (i.e., the sum of squared distances to the nearest cluster center).

labels	Age	INCOME	Wealth	Education	IncomevsGift	Donation_semester	MoneyPerYear	First_month	AVGGIFT	NGIFTALL	NUMCHLD
2	66.8	3.4	4.8	2.4	5.5	0.7	13.5	106.8	10.2	12.4	0.1
1	47.9	5.3	6.3	2.7	3.6	0.7	18.4	47.0	14.6	5.2	0.4
4	62.8	2.8	4.4	2.3	9.9	0.7	19.8	32.9	14.9	4.0	0.1
0	61.4	3.5	4.7	2.4	10.9	1.6	27.3	97.4	8.8	24.4	0.2
3	59.0	4.3	5.8	2.7	27.7	1.1	86.5	52.4	46.4	9.6	0.1

Fig.9 Final clustering, using k-means with k=5

From figure 8, we can conclude that each cluster shows a different type of donor:

- **#2:** the donors which donated the longest time ago; eldest group; donated less money per year and have a relatively low income/gift (donation) ratio;
- **#1:** youngest group of donors; lowest income/gift ratio; highest income and wealth;
- **#4:** donors which started donating more recently; they donated per year, on average, $1.4 more than cluster #1, however they are the ones with the lowest income; lowest education rating.
- **#0:** cluster which contributed more times during the year but have the lowest dollar average per gift. In total, they donated two times more than cluster #0, #1 and #2.
- **#3:** highest average donation contribution; highest amount of money donated per year and highest income/gift ratio: 27,7. Most educated group, ex-aequo, with group #1.

4.2 GMM

GMM stands for Gaussian Mixture Models (see reference [9]) and it can be used, like k-means, to cluster unlabeled data. GMM has, however, two main advantages over k-means:

1) GMM accounts for variance and k-means doesn’t, which means that GMM can adapt better to non-circular shaped clusters;
2) The second difference between k-means and Gaussian mixture models is that the former performs hard classification whereas the latter performs soft classification. In other words, k-means tells us what data point belongs to which cluster but won’t provide us with the probabilities that a given data point belongs to each of the possible clusters.

With this in mind, we decided to experiment with a GMM model to cluster our data but concluded that GMM shows a lower discriminatory power than k-means (see figure 9): the top graph shows GMM clustering while the bottom one shows k-means clustering.

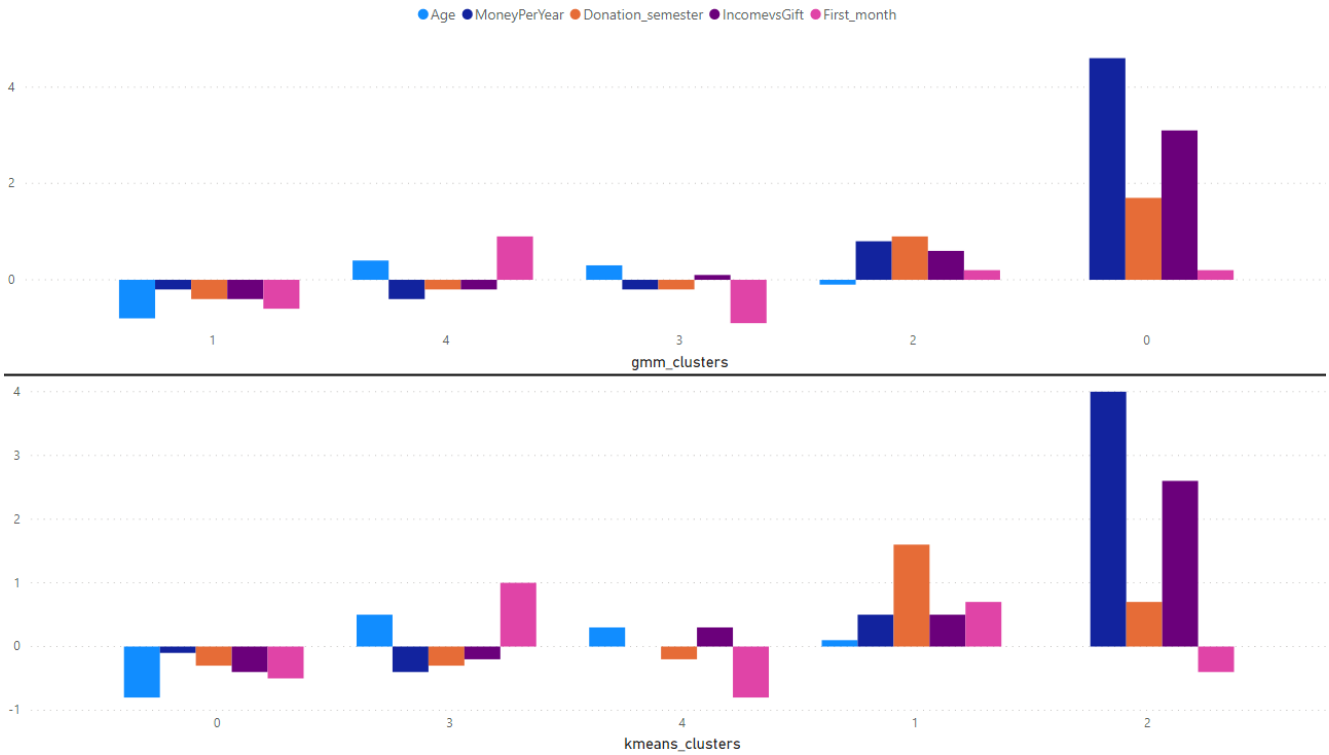

Fig.10 GMM vs K-means clustering

Each barplot represents a cluster for a total of 5 clusters. After scaling the data to be easily comparable, you can observe how each variable, on average, behaves in each cluster. You can observe how k-means shows a slightly better variance in comparison to GMM, which will help us to have a clearer distinction between clusters, which donor is in each cluster and the appropriate marketing strategy for each group. For that reason, we have decided to keep k-means as our data segmentation model.

4.3 t-SNE

To better visualize our clustering approach we went through both the barplot showed before, a spiderweb diagram (in figure 10), and finally, t-SNE.

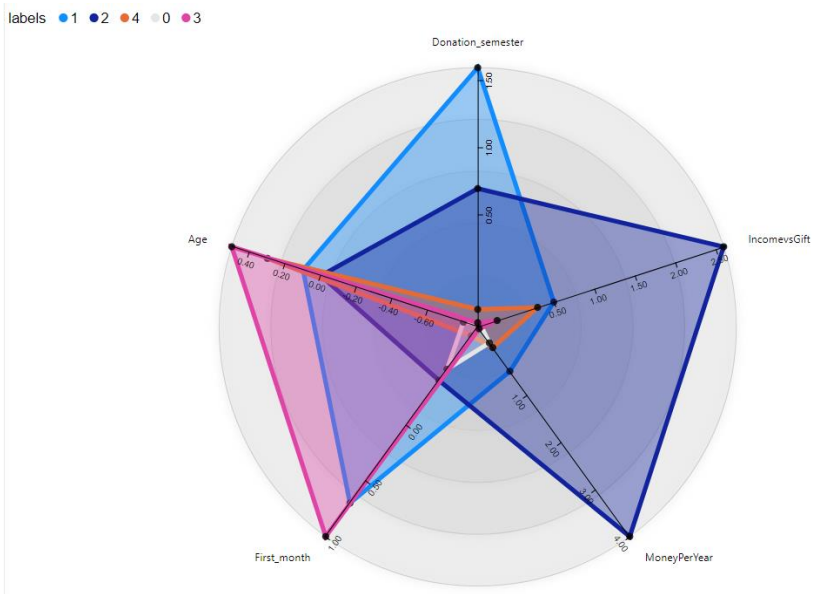

Fig.11 Spiderweb diagram

Since we are in a high dimensional space we decided to go through t-SNE a probability-based approach to transform our data into a two-dimensional space based on the probability of two observations to be close to each other. Various learning rates have been tried but the most suitable one is the default one, 200. In few words, t-sne tries to capture local and global structure of high-dimensional data. It tries to misnimize the difference between the probability of two observations to be close in the high-dimensional space against the probability of the corresponding points on a 2 dimensional space; this assuming that both probability follow the same distribution.

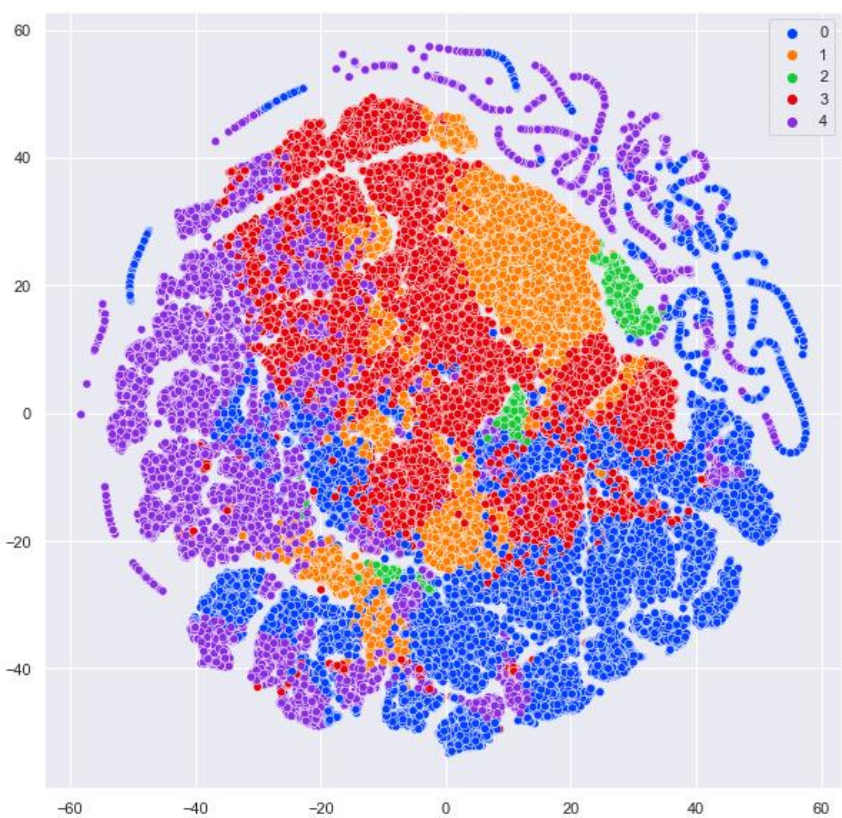

Fig.12 t-SNE visualization

4.4 K-Prototype

From our literature review, multiple articles suggested the use of K-Prototype for mixed data types [10] [11] [12]. This is tool is similar to K-means approach in that it utilizes Euclidean distance for numerical variables, but also uses the number of matching categories to measure the distance between categorical variables [13]. Such treatment of categorical variables adds an advantage in that it minimizes the dilution of the context of each level of the categorical feature. A simplified example: consider the variable Wealth. A half of Cluster A may be Wealth = 3, and the other half 5. In K-means, this would be represented as Cluster A mostly consists of individuals with Wealth level of 4, the average of the two. K-Prototype, instead, assigns a cluster to each individual and allows us to study their nature using various statistical analysis, such as median, mode, standard deviation etc.

For this clusterization, we selected the following features: *Urbanicity*, *Age*, *GENDER*, *Wealth*, *Education*, *IC1*, *NUMCHLD*, and *MoneyPerYear*. INCOME variable was rejected, as we noticed many of the donors are around the age of retirement. These individuals may have high level of saving, but low income level, if not zero. The numerical variables (Age, IC1, NUMCHLD, and MoneyPerYear) were first normalized with standard scaling. The categorical variables do not require such preprocessing. We settled on the number of clusters = 3, which was rather an arbitrary number. N_clusters = 2 was rejected as it is too few. Four clusters resulted in very high concentration on Cluster 0 (70k observations) and Cluster 3 with only 15 observations. Furthermore, four or more clusters we tested resulted in very low distinction among the clusters.

	Urbanicity	*Age*	*GENDER*	*Wealth*	*Education*	*IC1*	*NUMCHLD*	*Money PerYear*
Cluster 0, 47716 Observations (18.6%)								
Mean	2.588	59.353	0.454	7.971	3.313	588.674	0.169	23.728
Mode	2	57	0	9	3	518	0	15.000
Median	2.0	59.0	0.0	8.0	3.0	546.0	0.0	18.000
75%	3	68	1	9	4	641	0	25.000
Cluster 1, 26293 Observations (52.0%)								
Mean	2.650	68.698	0.427	4.092	2.229	272.886	0.050	18.884
Mode	2	61	0	3	2	213	0	15.000
Median	2.0	67.0	0.0	4.0	2.0	267.0	0.0	15.519
75%	3	76	1	6	2	332	0	22.392
Cluster 2, 16864 Observations (28.2%)								
Mean	2.655	42.355	0.414	4.864	2.334	311.477	0.507	19.463
Mode	2	47	0	7	2	263	0	15.000
Median	2.0	44.0	0.0	5.0	2.0	308.0	0.0	16.421
75%	3	49	1	7	3	383	1	23.000

Fig.13 Summary of characteristics used for analysis, for each cluster.

Cluster 0 is clearly the wealthiest in our analysis, as shown in Figure 13. It is also the individuals who made the highest amount of donation per year. It is important to note that Wealth variable is relative to the resident state. Meaning, we can only conclude these individuals who are wealthier than the surrounding neighborhood tends to make higher amount of donation to PVA. We cannot conclude whether the wealthy population in absolute term makes higher contribution with the data provided to us. These individuals also reside in highly education neighborhood (considering the census data is actually from 1990, when the university degree holder was rare).

The difference in the mean yearly contribution amount of Cluster 2 is only slightly higher than Cluster 1, despite the significant difference in their age and the number of children. This finding, in addition to the fact that the mean Wealth level of Cluster 2 is slightly higher than Cluster 1, may indicate Wealth level has higher impact on the donation behaviors than other variables.

Urbanicity failed to produce a visible distinction among the clusters. Perhaps for future research it would be wise to adopt equal depth discretization for the variable. Additionally, in all clusters males are more likely to donate to PVA than females.

5. Marketing Recommendation

Now that we have defined our 5 clusters and the type of donor in each one of them, I shall call the attention of the reader for the initial purpose of this report: we pretended to segment the lapsed PVA donors (done) and create targeted marketing strategies for each group to recapture these former donors, which we will explore in this chapter.

Our first approach to this problem was to identify all the possible features in our dataset that gave us some preference information about the donors (if they were interested in the bible, if they were plate collectors, etc.). We quickly realized, however, that only about a third of the observations had marketing related data. If you take a look at the table below, as indexes we have our 5 clusters and the numbers in each of the columns represent the % of 1s (donors that are interested in the product) on the total of 1s for the cluster. For instance, 4.2% of cluster 0 replied 1 (are interested) to COLLECT1.

labels	COLLECT1	VETERANS	BIBLE	CATLG	HOMEE	PETS	CDPLAY	STEREO	PCOWNERS	PHOTO	CRAFTS	FISHER	GARDENIN	BOATS	WALKER
0	4.2	8.5	6.5	6.2	0.7	10.3	9.2	9.5	8.2	3.7	6.1	4.9	10.1	1.5	8.1
1	3.9	8.1	6.3	6.0	0.7	10.4	9.0	9.7	8.1	3.8	6.1	5.4	10.5	1.3	8.6
2	5.7	8.0	6.4	5.3	0.5	11.2	8.6	10.0	8.4	3.5	6.5	4.5	9.2	1.0	8.4
3	4.0	8.2	6.5	5.9	0.7	10.3	9.2	9.7	7.9	3.5	6.1	5.0	10.5	1.6	8.1
4	3.9	8.4	6.7	5.9	0.6	10.4	9.0	9.2	8.1	3.7	6.1	5.1	10.3	1.6	8.4

Fig.14 Donors interests percentages by cluster and marketing feature

From this snapshot, we quickly realize that this strategy won't help us draw any robust conclusions about the best marketing approach for each cluster: not only we are working with only a third of the total observations but we also don't have more than 11.2% of donors in any given cluster claiming their preference over a single product, which hardly makes it significant enough to scale it to the whole cluster.

Our second approach to the problem was to use our common sense and limited marketing knowledge to, with the information we had about each cluster, recommend possible marketing strategies to recapture these former donors:

- #2: This is the eldest group (66.8 years, on average). It is important to market in different platforms as many older people do not use email and social media often: announcing on newspapers or retirement homes may have a bigger effect. Since many people of these people are themselves veterans or personally know people that have been in war (in Vietnam for example) a marketing campaign that shared the message of "helping your brother in arms" could be successful.
- #1: This is the youngest group (47.9 years, on average) with the highest income. Most of them are actively working and on the height of their careers. The best time to recapture and monetize these donors attention

would be when it's time to declare taxes. At that moment, advertise heavily on social media and TV to try to capture part of their taxes as a donation for PVA.

- #4: These are the most recent donors. We need to keep them engaged while they remember who PVA is and what it does: send monthly newsletters with real veterans' testimonies about how PVA helps them; create PVA loyalty cards; incentivize shares on social media.
- #0: These donors do more donations, but they donate relatively little each time. Our goal with donors in this cluster is to increase their dollar donations average. For example, we could create a campaign specifically designed for these donors that if they donated more than x dollars per donation, they would win a gift card or a discount in one of PVA's partners/sponsors or even a ticket to a PVA's organized lottery.
- #3: Our VIP's, they clearly love PVA and what it represents! These are our biggest donors and we need to keep them happy and make sure that they bring more people in. Create a loyalty PVA card where these would be our "premium" members. The more money they donate and people they bring the higher they go on "PVA's hierarchy" which could give them access to discounts and even veterans' personalized thank you notes.

6. Conclusion

This report explores how data segmentation can be done using data mining techniques. Being given 95 412 donors with 476 features each, our first challenge was to reduce the number of features to the most significant ones (low number of missing values, coherent data, etc.). We decided to remove what we considered to be non-essential features: features that were redundant with another feature; attributes that were not meaningful to our project objectives; variables that were not clearly defined or when the definition in the metadata mismatched the data and also ambiguous features.

After this initial cut, we were left with 22 non-marketing features (15 original ones and 7 that we engineered). We then proceeded to clean the data to identify and remove errors and duplicates in the data, to create a reliable dataset (this was done through the elimination of outliers, missing values imputation, coherence chacks and scaling). We also performed PCA on our cleaned dataset to make the data mining process more efficient.

Finally, it was time to model our dataset and hopefully segment it in such a way that we could clearly distinguish specific donors characteristics and do marketing recommendations for each one of them. In these segments, we used a wide range of techniques such as k-means, GMM, t-SNE and k-prototype.

Our final segmentation strategy proposed, in which we performed our marketing recommendations to recapture these former donors, used both HC and k-means to segment the lapsed donors in 5 different groups.

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